ia_study1_computeMeasures

2026-03-11

Admin (Packages & Custom Functions)

```r
#setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.6.0
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Load Data

```r
data <- read_csv("ia_study1_anon_data.csv")
data <- data[ , -1]
```

Ten Item Personality Inventory (TIPI)

Next, we compute the composite scores of the participant-level TIPI variables. Specifically, we calculate:

1. participant's extraversion `tipi_ex`

- `tipi_1` (Extraverted, enthusiastic), `tipi_6` (reversed; Reserved, quiet)

2. participant's agreeableness, `tipi_a`

- `tipi_2` (reversed; Critical, quarrelsome), `tipi_7` (Sympathetic, warm)

3. participant's conscientiousness, `tipi_c`

- `tipi_3` (Dependable, self-disciplined), `tipi_8` (reversed; Disorganized, careless)

4. participant's emotoinal stability, `tipi_es`

- `tipi_4` (reversed; Anxious, easily upset), `tipi_9` (Calm, emotionally stable)

5. participant's openness to experience, `tipi_o`

- `tipi_5` (Open to new experiences, complex), `tipi_10` (reversed; Conventional, uncreative)

```r
# compute tipi variables
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # reverse code tipi items
    tipi_6r = 8 - tipi_6,
    tipi_2r = 8 - tipi_2,
    tipi_8r = 8 - tipi_8,
    tipi_4r = 8 - tipi_4,
    tipi_10r = 8 - tipi_10
    ) %>%
  dplyr::mutate( # calculate tipi_ex, tipi_a, tipi_c, tipi_es, and tipi_o
    tipi_ex = mean(c(tipi_1, tipi_6r), na.rm = T),
    tipi_a = mean(c(tipi_2r, tipi_7), na.rm = T),
    tipi_c = mean(c(tipi_3, tipi_8r), na.rm = T),
    tipi_es = mean(c(tipi_4r, tipi_9), na.rm = T),
    tipi_o = mean(c(tipi_5, tipi_10r), na.rm = T)
    ) %>%
  ungroup()
```

```r
data %>%
  dplyr::select("tipi_1", "tipi_6r",
                "tipi_2r", "tipi_7",
                "tipi_3", "tipi_8r",
                "tipi_4r", "tipi_9",
                "tipi_5", "tipi_10r") %>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##          tipi_1 tipi_6r tipi_2r tipi_7 tipi_3 tipi_8r tipi_4r tipi_9 tipi_5
## tipi_1     1.00    0.62   -0.23   0.10   0.04   -0.22   -0.01  -0.10   0.23
## tipi_6r    0.62    1.00   -0.09   0.02  -0.09   -0.15    0.04  -0.23   0.09
## tipi_2r   -0.23   -0.09    1.00   0.12  -0.04    0.26    0.14   0.08   0.02
## tipi_7     0.10    0.02    0.12   1.00   0.03    0.04   -0.15   0.00   0.15
## tipi_3     0.04   -0.09   -0.04   0.03   1.00    0.45    0.10   0.30   0.21
## tipi_8r   -0.22   -0.15    0.26   0.04   0.45    1.00    0.23   0.18  -0.07
## tipi_4r   -0.01    0.04    0.14  -0.15   0.10    0.23    1.00   0.43   0.00
## tipi_9    -0.10   -0.23    0.08   0.00   0.30    0.18    0.43   1.00   0.08
## tipi_5     0.23    0.09    0.02   0.15   0.21   -0.07    0.00   0.08   1.00
## tipi_10r   0.06    0.21    0.08   0.02   0.01    0.07    0.24  -0.14   0.27
##          tipi_10r
## tipi_1       0.06
## tipi_6r      0.21
## tipi_2r      0.08
## tipi_7       0.02
## tipi_3       0.01
## tipi_8r      0.07
## tipi_4r      0.24
## tipi_9      -0.14
## tipi_5       0.27
## tipi_10r     1.00
```

```r
# European American
data %>% dplyr::filter(cultureEA == 1) %>%
  dplyr::select("tipi_1", "tipi_6r",
                "tipi_2r", "tipi_7",
                "tipi_3", "tipi_8r",
                "tipi_4r", "tipi_9",
               "tipi_5", "tipi_10r") %>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##          tipi_1 tipi_6r tipi_2r tipi_7 tipi_3 tipi_8r tipi_4r tipi_9 tipi_5
## tipi_1     1.00    0.58   -0.19   0.20   0.07   -0.16    0.09   0.14   0.24
## tipi_6r    0.58    1.00    0.02  -0.03  -0.05    0.08    0.22  -0.07   0.11
## tipi_2r   -0.19    0.02    1.00   0.14  -0.16    0.18    0.16  -0.18   0.03
## tipi_7     0.20   -0.03    0.14   1.00   0.09    0.01   -0.24  -0.05   0.10
## tipi_3     0.07   -0.05   -0.16   0.09   1.00    0.36    0.12   0.41   0.05
## tipi_8r   -0.16    0.08    0.18   0.01   0.36    1.00    0.31   0.13  -0.17
## tipi_4r    0.09    0.22    0.16  -0.24   0.12    0.31    1.00   0.48  -0.01
## tipi_9     0.14   -0.07   -0.18  -0.05   0.41    0.13    0.48   1.00   0.11
## tipi_5     0.24    0.11    0.03   0.10   0.05   -0.17   -0.01   0.11   1.00
## tipi_10r  -0.04    0.22    0.25   0.03  -0.16    0.23    0.11  -0.26   0.18
##          tipi_10r
## tipi_1      -0.04
## tipi_6r      0.22
## tipi_2r      0.25
## tipi_7       0.03
## tipi_3      -0.16
## tipi_8r      0.23
## tipi_4r      0.11
## tipi_9      -0.26
## tipi_5       0.18
```

```
## tipi_10r     1.00
```

```r
# Taiwanese
data %>% dplyr::filter(cultureEA == 0) %>%
  dplyr::select("tipi_1", "tipi_6r",
                "tipi_2r", "tipi_7",
                "tipi_3", "tipi_8r",
                "tipi_4r", "tipi_9",
                "tipi_5", "tipi_10r")%>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##          tipi_1 tipi_6r tipi_2r tipi_7 tipi_3 tipi_8r tipi_4r tipi_9 tipi_5
## tipi_1     1.00    0.68   -0.28   0.00   0.02   -0.31   -0.13  -0.39   0.21
## tipi_6r    0.68    1.00   -0.18   0.07  -0.06   -0.32   -0.10  -0.39   0.11
## tipi_2r   -0.28   -0.18    1.00   0.10   0.08    0.37    0.14   0.32   0.02
## tipi_7     0.00    0.07    0.10   1.00  -0.05    0.06   -0.07   0.04   0.18
## tipi_3     0.02   -0.06    0.08  -0.05   1.00    0.42    0.00   0.27   0.28
## tipi_8r   -0.31   -0.32    0.37   0.06   0.42    1.00    0.10   0.26  -0.05
## tipi_4r   -0.13   -0.10    0.14  -0.07   0.00    0.10    1.00   0.38  -0.03
## tipi_9    -0.39   -0.39    0.32   0.04   0.27    0.26    0.38   1.00   0.06
## tipi_5     0.21    0.11    0.02   0.18   0.28   -0.05   -0.03   0.06   1.00
## tipi_10r   0.17    0.26   -0.02  -0.01  -0.02   -0.18    0.31  -0.03   0.31
##          tipi_10r
## tipi_1       0.17
## tipi_6r      0.26
## tipi_2r     -0.02
## tipi_7      -0.01
## tipi_3      -0.02
## tipi_8r     -0.18
## tipi_4r      0.31
## tipi_9      -0.03
## tipi_5       0.31
## tipi_10r     1.00
```

Interpersonal Reativity Index (IRI)

Next, we compute the composite scores of the participant-level IRI variables. Specifically, we calculate:

1. participant's perspective-taking `iri_pt`

- calculated by averaging `iri3`, `iri8`, `iri11`, `iri15`, `iri21`, `iri25`, & `iri28`

2. participant's fantasy, `iri_fs`

- calculated by averaging `iri1`, `iri5`, `iri7`, `iri12`, `iri16`, `iri23`, & `iri26`

3. participant's empathetic concern, `iri_ec`

- calculated by averaging `iri2`, `iri4`, `iri9`, `iri14`, `iri18`, `iri20`, & `iri22`

4. participant's personal distress, `iri_pd`

- calculated by averaging `iri6`, `iri10`, `iri13`, `iri17`, `iri19`, `iri24`, & `iri27`

```
# compute iri variables
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # reverse code iri items
    iri3r = 6 - iri3,
    iri4r = 6 - iri4,
    iri7r = 6 - iri7,
    iri12r = 6 - iri12,
    iri13r = 6 - iri13,
    iri14r = 6 - iri14,
    iri15r = 6 - iri15,
    iri18r = 6 - iri18,
    iri19r = 6 - iri19,
    ) %>%
  dplyr::mutate( # calculate iri_pt, iri_fs, iri_ec, iri_pd
    iri_pt = mean(
      c(iri3r, iri8, iri11, iri15r, iri21, iri25, iri28), na.rm = T),
    iri_fs = mean(
      c(iri1, iri5, iri7r, iri12r, iri16, iri23, iri26), na.rm = T),
    iri_ec = mean(
      c(iri2, iri4r, iri9, iri14r, iri18r, iri20, iri22), na.rm = T),
    iri_pd = mean(
      c(iri6, iri10, iri13r, iri17, iri19r, iri24, iri27), na.rm = T)
    ) %>%
  ungroup()
```

```
reliab_iri <- c("iri3r, iri8, iri11, iri15r, iri21, iri25, iri28",
               "iri1, iri5, iri7r, iri12r, iri16, iri23, iri26",
               "iri2, iri4r, iri9, iri14r, iri18r, iri20, iri22",
               "iri6, iri10, iri13r, iri17, iri19r, iri24, iri27")
```

```r
reliab_iri_names <- c("iri_pt", "iri_fs", "iri_ec", "iri_pd")

all_alphas(data, items = reliab_iri, constructs = reliab_iri_names,
           sensitivity = sensitivity_analyses)
```

```
## # A tibble: 12 x 5
##    construct      `!is.na(subID)` `exclude_age != 1` `exclude_sd != 1`
##    <chr>                    <dbl>              <dbl>             <dbl>
##  1 iri_pt_overall            0.77               0.77              0.77
##  2 iri_fs_overall            0.8                0.8               0.78
##  3 iri_ec_overall            0.79               0.8               0.79
##  4 iri_pd_overall            0.8                0.8               0.79
##  5 iri_pt_EA                 0.78               0.79              0.78
##  6 iri_fs_EA                 0.83               0.84              0.82
##  7 iri_ec_EA                 0.78               0.8               0.78
##  8 iri_pd_EA                 0.84               0.84              0.84
##  9 iri_pt_TN                 0.76               0.76              0.76
## 10 iri_fs_TN                 0.76               0.76              0.76
## 11 iri_ec_TN                 0.79               0.8               0.79
## 12 iri_pd_TN                 0.75               0.75              0.71
## # i 1 more variable: `exclude_age != 1 & exclude_sd != 1` <dbl>
```

AVI - Ideal and Actual HAP & LAP

Now, we compute the composite scores of the participant-level AVI variables. Specifically, we calculate:

1. participant's actual HAP, `rHAP`

- calculated by averaging `r1_1` (enthusiastic), `r1_7` (excited), `r1_9` (elated), `r2_9` (euphoric)

2. participant's actual LAP, `rLAP`

- calculated by averaging `r1_6` (relaxed), `r3_3` (calm), `r3_9` (peaceful), `r3_10` (serene)

3. participant's ideal HAP, `iHAP`

- calculated by averaging `i1_1` (enthusiastic), `i1_7` (excited), `i1_9` (elated), `i2_9` (euphoric)

4. participant's ideal LAP, `iLAP`

- calculated by averaging `i1_6` (relaxed), `i3_3` (calm), `i3_9` (peaceful), `i3_10` (serene)

5. participant's ipsatized `rHAP`, `rLAP`, `iHAP`, and `iLAP` scores

- calculated by converting ratings using this formula: `ipsatized score = (raw score - individuals' mean ratings across all emotion items) / (individuals' sd across the ratings on all emotion items)`

```r
# compute AVI variables
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # create rHAP, rLAP, iHAP, iLAP
    rHAP = mean(c(r1_1, r1_7, r1_9, r2_9), na.rm = T),
    rLAP = mean(c(r1_6, r3_3, r3_9, r3_10), na.rm = T),
    iHAP = mean(c(i1_1, i1_7, i1_9, i2_9), na.rm = T),
    iLAP = mean(c(i1_6, i3_3, i3_9, i3_10), na.rm = T)) %>%
  dplyr::mutate( # create intermediate variables for ipsatize scores
    rMean = mean(c_across(matches("^r[1-4]_[1-9][0]*$")), na.rm = TRUE),
    rSd = sd(c_across(matches("^r[1-4]_[1-9]$|^r4_10$")), na.rm = TRUE),
    iMean = mean(c_across(matches("^i[1-4]_[1-9][0]*$")), na.rm = TRUE),
    iSd = sd(c_across(matches("^i[1-4]_[1-9]$|^i4_10$")), na.rm = TRUE)) %>%
  dplyr::mutate( # create ipsatized item scores
    across(matches("^r[1-4]_[1-9][0]*$"),
           ~ as.numeric((.x - rMean) / rSd),
           .names = "{.col}_i"),
    across(matches("^i[1-4]_[1-9][0]*$"),
           ~ as.numeric((.x - iMean) / iSd),
           .names = "{.col}_i")) %>%
  dplyr::mutate( # create ipsatized rHAP, rLAP, iHAP, iLAP
    rHAP_i = mean(c(r1_1_i, r1_7_i, r1_9_i, r2_9_i), na.rm = T),
    rLAP_i = mean(c(r1_6_i, r3_3_i, r3_9_i, r3_10_i), na.rm = T),
    iHAP_i = mean(c(i1_1_i, i1_7_i, i1_9_i, i2_9_i), na.rm = T),
    iLAP_i = mean(c(i1_6_i, i3_3_i, i3_9_i, i3_10_i), na.rm = T)) %>%
  ungroup()
```

```r
# calculate cronbach alpha

reliab_avi <- c("r1_1, r1_7, r1_9, r2_9",
                "r1_6, r3_3, r3_9, r3_10",
                "i1_1, i1_7, i1_9, i2_9",
                "i1_6, i3_3, i3_9, i3_10")

reliab_avi_names <- c("rHAP", "rLAP", "iHAP", "iLAP")

all_alphas(data, items = reliab_avi, constructs = reliab_avi_names,
           sensitivity = sensitivity_analyses)
```

```
## # A tibble: 12 x 5
##    construct    `!is.na(subID)` `exclude_age != 1` `exclude_sd != 1`
##    <chr>                  <dbl>              <dbl>             <dbl>
##  1 rHAP_overall            0.83               0.81              0.8
##  2 rLAP_overall            0.85               0.85              0.84
##  3 iHAP_overall            0.79               0.79              0.76
##  4 iLAP_overall            0.76               0.77              0.75
##  5 rHAP_EA                 0.87               0.85              0.84
##  6 rLAP_EA                 0.85               0.86              0.83
##  7 iHAP_EA                 0.8                0.79              0.74
##  8 iLAP_EA                 0.73               0.75              0.69
##  9 rHAP_TN                 0.78               0.78              0.77
## 10 rLAP_TN                 0.86               0.86              0.85
## 11 iHAP_TN                 0.77               0.77              0.77
## 12 iLAP_TN                 0.81               0.81              0.81
## # i 1 more variable: `exclude_age != 1 & exclude_sd != 1` <dbl>
```

AVI - Maximizing Positivity

Now, we compute the composite scores of the participant-level AVI variables. Specifically, we calculate:

1. participant's actual PA, `rPA`

- calculated by averaging `r1_1` (enthusiastic), `r1_7` (excited), `r1_9` (elated), `r2_9` (euphoric), `r1_6` (relaxed), `r3_3` (calm), `r3_9` (peaceful), `r3_10` (serene), `r3_1` (happy), `r3_7` (satisfied), `r2_5` (content)

2. participant's actual NA, `rNA`

- calculated by averaging `r2_10` (fearful), `r3_6` (hostile), `r1_3` (nervous), `r2_8` (sad), `r2_2` (lonely), `r3_4` (unhappy), `r2_6` (sluggish), `r1_10` (sleepy), `r1_4` (dull)

3. participant's ideal PA, `iPA`

- calculated by averaging `i1_1` (enthusiastic), `i1_7` (excited), `i1_9` (elated), `i2_9` (euphoric), `i1_6` (relaxed), `i3_3` (calm), `i3_9` (peaceful), `i3_10` (serene), `i3_1` (happy), `i3_7` (satisfied), `i2_5` (content)

4. participant's ideal NA, `iNA`

- calculated by averaging `i2_10` (fearful), `i3_6` (hostile), `i1_3` (nervous), `i2_8` (sad), `i2_2` (lonely), `i3_4` (unhappy), `i2_6` (sluggish), `i1_10` (sleepy), `i1_4` (dull)

5. participant's ipsatized `rPA`, `rNA`, `iPA`, and `iNA` scores
6. participant's raw maximizing positivity scores `maxPos` and ipsatized maximizing positivity scores `maxPos_i`

- calculated by subtracting (ipsatized) ideal NA from ideal PA

```
# compute AVI variables
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # create rPA, rNA, iPA, iNA
    rPA = mean(
      c(r1_1, r1_7, r1_9, r2_9, r1_6, r3_3, r3_9, r3_10, r3_1, r3_7, r2_5),
      na.rm = T),
    rNA = mean(
      c(r2_10, r3_6, r1_3, r2_8, r2_2, r3_4, r2_6, r1_10, r1_4),
      na.rm = T),
    iPA = mean(
      c(i1_1, i1_7, i1_9, i2_9, i1_6, i3_3, i3_9, i3_10, i3_1, i3_7, i2_5),
      na.rm = T),
    iNA = mean(
      c(i2_10, i3_6, i1_3, i2_8, i2_2, i3_4, i2_6, i1_10, i1_4),
      na.rm = T),

    rPA_i = mean(
      c(r1_1_i, r1_7_i, r1_9_i, r2_9_i, r1_6_i, r3_3_i, r3_9_i, r3_10_i,
        r3_1_i, r3_7_i, r2_5_i), na.rm = T),
    rNA_i = mean(
```

```r
      c(r2_10_i, r3_6_i, r1_3_i, r2_8_i, r2_2_i, r3_4_i, r2_6_i, r1_10_i,
        r1_4_i), na.rm = T),
    iPA_i = mean(
      c(i1_1_i, i1_7_i, i1_9_i, i2_9_i, i1_6_i, i3_3_i, i3_9_i, i3_10_i,
        i3_1_i, i3_7_i, i2_5_i), na.rm = T),
    iNA_i = mean(
      c(i2_10_i, i3_6_i, i1_3_i, i2_8_i, i2_2_i, i3_4_i, i2_6_i, i1_10_i,
        i1_4_i), na.rm = T)) %>%
  dplyr::mutate(
    # create maxPos variables
    maxPos = iPA - iNA,
    maxPos_i = iPA_i - iNA_i) %>%
  ungroup()
```

```r
# calculate cronbach alpha

reliab_avimp <-
  c("r1_1, r1_7, r1_9, r2_9, r1_6, r3_3, r3_9, r3_10, r3_1, r3_7, r2_5",
    "r2_10, r3_6, r1_3, r2_8, r2_2, r3_4, r2_6, r1_10, r1_4",
    "i1_1, i1_7, i1_9, i2_9, i1_6, i3_3, i3_9, i3_10, i3_1, i3_7, i2_5",
    "i2_10, i3_6, i1_3, i2_8, i2_2, i3_4, i2_6, i1_10, i1_4")

reliab_avi_namesmp <- c("rPA", "rNA", "iPA", "iNA")

all_alphas(data, items = reliab_avimp, constructs = reliab_avi_namesmp,
           sensitivity = sensitivity_analyses)
```

```
## # A tibble: 12 x 5
##    construct   `!is.na(subID)` `exclude_age != 1` `exclude_sd != 1`
##    <chr>                 <dbl>              <dbl>             <dbl>
##  1 rPA_overall            0.87               0.87              0.86
##  2 rNA_overall            0.86               0.86              0.85
##  3 iPA_overall            0.84               0.84              0.84
##  4 iNA_overall            0.88               0.88              0.83
##  5 rPA_EA                 0.9                0.9               0.89
##  6 rNA_EA                 0.89               0.9               0.89
##  7 iPA_EA                 0.84               0.85              0.83
##  8 iNA_EA                 0.93               0.94              0.89
##  9 rPA_TN                 0.85               0.85              0.84
## 10 rNA_TN                 0.83               0.83              0.83
## 11 iPA_TN                 0.85               0.85              0.86
## 12 iNA_TN                 0.77               0.76              0.77
## # i 1 more variable: `exclude_age != 1 & exclude_sd != 1` <dbl>
```

Ipsatized giving scores

Now, we compute the ipsatized scores of the trials. Alike the AVI variables, we calculate this by converting the amount of money given using this formula: `ipsatized score = (raw amoung given - individuals' mean amount given across all trials) / (individuals' sd across the amount of money given across all trials)`

```
# compute ipsatized giving scores
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # create intermediate variables for calculating ipsatize scores
    givingMean = mean(
      c_across(matches("^raw[1-9][1-9]*[0]*$")), na.rm = TRUE),
    givingSd = sd(
      c_across(matches("^raw[1-9][1-9]*[0]*$")), na.rm = TRUE)) %>%
  dplyr::mutate( # create ipsatized item scores
    across(matches("^raw[1-9][1-9]*[0]*$"),
           ~ as.numeric((.x - givingMean) / givingSd),
           .names = "{.col}_i")) %>%
  ungroup()
```

Ipsatized trait rating scores

Finally, we compute the ipsatized scores of the trait ratings for the faces (items that start with `trait`, `vale`, and `arou`).

All participants rated the faces on numerous traits. Specifically, they responded to the prompt "To what extent is this person…" (e.g., "Competent", "Trustworthy", etc.; 1 = *Not at all*, 4 = *Moderately*, 7 = *Extremely*) and how positive ("*How negative - positive do you feel when you see this face?*"; 1 = *Extremely negative*, 7 = *Extremely positive*) and aroused ("*How not-aroused - aroused do you feel when you see this face? Aroused means activated, charged, alert, or energized physically and mentally.*"; 1 = *Not at all aroused*, 7 = *Extremely aroused*) they feel when they see the face. For the repeated ratings, we only include the first rating they provide.

Alike the AVI and giving variables, we calculate this by converting the ratings given using this formula: `ipsatized score = (raw rating - individuals' mean rating across all traits and faces) / (individuals' sd rating across all traits and faces)`. Note that for valence (`vale`) and arousal (`arou`) ratings, they are ipsatized within the valence and arousal ratings only (i.e., `ipsatized valence = (raw valence rating - individuals' mean rating of valence across all faces) / (individuals' sd of valence across all faces)`.

```r
# compute ipsatized trait rating scores
data <- data %>%
  rowwise() %>%
  dplyr::mutate( # create intermediate variables for calculating ipsatize scores
    traitRatingMean = mean(
      c_across(matches("^trait[1-9][0-9]*_")), na.rm = TRUE),
    traitRatingSd = sd(
      c_across(matches("^trait[1-9][0-9]*_")), na.rm = TRUE),
    valeRatingMean = mean(
      c_across(matches("^vale[1-9][0-9]*$")), na.rm = TRUE),
    valeRatingSd = sd(
      c_across(matches("^vale[1-9][0-9]*$")), na.rm = TRUE),
    arouRatingMean = mean(
      c_across(matches("^arou[1-9][0-9]*$")), na.rm = TRUE),
    arouRatingSd = sd(
      c_across(matches("^arou[1-9][0-9]*$")), na.rm = TRUE)) %>%
  dplyr::mutate(
    # create ipsatized item scores
    across(matches("^trait[1-9][0-9]*_"),
           ~ as.numeric((.x - traitRatingMean) / traitRatingSd),
           .names = "{.col}_i")) %>%
  dplyr::mutate( # create ipsatized item scores
    across(matches("^vale[1-9][0-9]*$"),
           ~ as.numeric((.x - valeRatingMean) / valeRatingSd),
           .names = "{.col}_i")) %>%
  dplyr::mutate( # create ipsatized item scores
    across(matches("^arou[1-9][0-9]*$"),
           ~ as.numeric((.x - arouRatingMean) / arouRatingSd),
           .names = "{.col}_i")) %>%
  ungroup()
```

Save data computed frame

```
write.csv(data, "ia_study1_anon_data_computed.csv")
```

Descriptives

```
# descriptives function
describe_all <- function(data, filter_condition1) {
  for (i in 1:length(filter_condition1)) {
      subset_data <- data %>%
        filter(eval(parse(text = filter_condition1[i])))
   desc <- subset_data  %>% rowwise() %>%
    dplyr::mutate(
      avg_giving_amt = mean(
        c_across((starts_with("raw") & !contains("_Page.Submit")))),
      avg_giving_RT = mean(
        c_across((starts_with("raw") & contains("_Page.Submit"))))) %>%
      dplyr::select(gender_nonfemale, sexualpref_nonhetero, ethn_nonmajority,
                    age, famises2_numbered, selfses2_numbered,
                    tipi_ex, tipi_a, tipi_c, tipi_es, tipi_o,
                    iri_pt, iri_fs, iri_ec, iri_pd,
                    avg_giving_amt, avg_giving_RT,
                    rHAP, rLAP, iHAP, iLAP) %>%
     psych::describe(fast = TRUE, skew = FALSE)
  print(filter_condition1[i])
  print(desc)
  }
}
```

```r
# Pooled data
describe_all(data = data, sensitivity_analyses)
```

```
## [1] "!is.na(subID)"
##                      vars   n  mean    sd median   min    max  range   se
## gender_nonfemale        1 220  0.40  0.49   0.00  0.00   1.00   1.00 0.03
## sexualpref_nonhetero    2 220  0.23  0.42   0.00  0.00   1.00   1.00 0.03
## ethn_nonmajority        3 220  0.05  0.23   0.00  0.00   1.00   1.00 0.02
## age                     4 219 23.41  5.59  22.00 18.00  66.00  48.00 0.38
## famises2_numbered       5 219  4.90  2.16   5.00  1.00   8.00   7.00 0.15
## selfses2_numbered       6 216  2.24  1.96   1.00  1.00   8.00   7.00 0.13
## tipi_ex                 7 220  3.69  1.67   4.00  1.00   7.00   6.00 0.11
## tipi_a                  8 220  4.85  1.20   4.50  2.00   7.00   5.00 0.08
## tipi_c                  9 220  4.90  1.31   5.00  2.00   7.00   5.00 0.09
## tipi_es                10 220  3.95  1.39   4.00  1.00   7.00   6.00 0.09
## tipi_o                 11 220  5.00  1.13   5.00  1.50   7.00   5.50 0.08
## iri_pt                 12 220  3.58  0.68   3.57  1.29   5.00   3.71 0.05
## iri_fs                 13 220  3.57  0.80   3.64  1.00   5.00   4.00 0.05
## iri_ec                 14 220  3.65  0.75   3.71  1.29   5.00   3.71 0.05
## iri_pd                 15 220  2.92  0.78   3.00  1.00   4.71   3.71 0.05
## avg_giving_amt         16 184  0.98  0.53   0.99  0.03   2.47   2.44 0.04
## avg_giving_RT          17 220  7.42 20.10   4.46  2.05 234.63 232.58 1.36
## rHAP                   18 220  2.36  0.69   2.25  1.00   5.00   4.00 0.05
## rLAP                   19 220  2.83  0.80   2.75  1.00   5.00   4.00 0.05
## iHAP                   20 220  3.14  0.78   3.00  1.00   5.00   4.00 0.05
## iLAP                   21 220  3.95  0.71   4.00  2.00   5.00   3.00 0.05
## [1] "exclude_age != 1"
##                      vars   n  mean    sd median   min    max  range   se
## gender_nonfemale        1 206  0.40  0.49   0.00  0.00   1.00   1.00 0.03
## sexualpref_nonhetero    2 206  0.23  0.42   0.00  0.00   1.00   1.00 0.03
## ethn_nonmajority        3 206  0.06  0.23   0.00  0.00   1.00   1.00 0.02
## age                     4 206 22.48  3.46  22.00 18.00  30.00  12.00 0.24
## famises2_numbered       5 205  4.83  2.17   5.00  1.00   8.00   7.00 0.15
## selfses2_numbered       6 202  2.07  1.82   1.00  1.00   8.00   7.00 0.13
## tipi_ex                 7 206  3.67  1.67   3.75  1.00   7.00   6.00 0.12
## tipi_a                  8 206  4.86  1.20   4.50  2.00   7.00   5.00 0.08
## tipi_c                  9 206  4.90  1.32   5.00  2.00   7.00   5.00 0.09
## tipi_es                10 206  3.94  1.38   4.00  1.00   7.00   6.00 0.10
## tipi_o                 11 206  4.97  1.12   5.00  1.50   7.00   5.50 0.08
## iri_pt                 12 206  3.58  0.68   3.57  1.29   5.00   3.71 0.05
## iri_fs                 13 206  3.59  0.80   3.71  1.00   5.00   4.00 0.06
## iri_ec                 14 206  3.65  0.76   3.71  1.29   5.00   3.71 0.05
## iri_pd                 15 206  2.95  0.76   3.00  1.00   4.71   3.71 0.05
## avg_giving_amt         16 175  0.98  0.53   0.99  0.03   2.47   2.44 0.04
## avg_giving_RT          17 206  6.72 17.55   4.48  2.05 234.63 232.58 1.22
## rHAP                   18 206  2.33  0.66   2.25  1.00   5.00   4.00 0.05
## rLAP                   19 206  2.84  0.82   2.75  1.00   5.00   4.00 0.06
## iHAP                   20 206  3.14  0.77   3.00  1.25   5.00   3.75 0.05
## iLAP                   21 206  3.96  0.72   4.00  2.00   5.00   3.00 0.05
## [1] "exclude_sd != 1"
##                      vars   n  mean    sd median   min    max  range   se
## gender_nonfemale        1 184  0.40  0.49   0.00  0.00   1.00   1.00 0.04
## sexualpref_nonhetero    2 184  0.24  0.43   0.00  0.00   1.00   1.00 0.03
```

```
## ethn_nonmajority        3 184  0.05  0.23   0.00  0.00   1.00   1.00 0.02
## age                     4 183 22.67  4.12  22.00 18.00  43.00  25.00 0.30
## famises2_numbered       5 183  4.69  2.13   5.00  1.00   8.00   7.00 0.16
## selfses2_numbered       6 180  2.09  1.82   1.00  1.00   8.00   7.00 0.14
## tipi_ex                 7 184  3.77  1.60   4.00  1.00   7.00   6.00 0.12
## tipi_a                  8 184  4.85  1.19   4.50  2.00   7.00   5.00 0.09
## tipi_c                  9 184  4.83  1.30   5.00  2.00   7.00   5.00 0.10
## tipi_es                10 184  3.89  1.31   4.00  1.00   7.00   6.00 0.10
## tipi_o                 11 184  4.99  1.09   5.00  1.50   7.00   5.50 0.08
## iri_pt                 12 184  3.59  0.67   3.71  1.29   5.00   3.71 0.05
## iri_fs                 13 184  3.59  0.77   3.57  1.00   5.00   4.00 0.06
## iri_ec                 14 184  3.66  0.73   3.71  1.29   5.00   3.71 0.05
## iri_pd                 15 184  2.94  0.73   3.00  1.00   4.71   3.71 0.05
## avg_giving_amt         16 184  0.98  0.53   0.99  0.03   2.47   2.44 0.04
## avg_giving_RT          17 184  6.65 17.60   4.73  2.13 234.63 232.50 1.30
## rHAP                   18 184  2.40  0.66   2.25  1.00   4.25   3.25 0.05
## rLAP                   19 184  2.86  0.78   2.75  1.00   5.00   4.00 0.06
## iHAP                   20 184  3.17  0.74   3.00  1.25   5.00   3.75 0.05
## iLAP                   21 184  3.95  0.70   4.00  2.00   5.00   3.00 0.05
## [1] "exclude_age != 1 & exclude_sd != 1"
##                      vars   n  mean    sd median   min    max  range   se
## gender_nonfemale        1 175  0.40  0.49   0.00  0.00   1.00   1.00 0.04
## sexualpref_nonhetero    2 175  0.24  0.43   0.00  0.00   1.00   1.00 0.03
## ethn_nonmajority        3 175  0.06  0.23   0.00  0.00   1.00   1.00 0.02
## age                     4 175 22.16  3.34  21.00 18.00  30.00  12.00 0.25
## famises2_numbered       5 174  4.63  2.13   4.50  1.00   8.00   7.00 0.16
## selfses2_numbered       6 171  1.95  1.70   1.00  1.00   8.00   7.00 0.13
## tipi_ex                 7 175  3.74  1.61   4.00  1.00   7.00   6.00 0.12
## tipi_a                  8 175  4.87  1.20   4.50  2.00   7.00   5.00 0.09
## tipi_c                  9 175  4.84  1.31   5.00  2.00   7.00   5.00 0.10
## tipi_es                10 175  3.89  1.33   4.00  1.00   7.00   6.00 0.10
## tipi_o                 11 175  4.97  1.10   5.00  1.50   7.00   5.50 0.08
## iri_pt                 12 175  3.60  0.66   3.71  1.29   5.00   3.71 0.05
## iri_fs                 13 175  3.60  0.78   3.71  1.00   5.00   4.00 0.06
## iri_ec                 14 175  3.66  0.74   3.71  1.29   5.00   3.71 0.06
## iri_pd                 15 175  2.95  0.73   3.00  1.00   4.71   3.71 0.06
## avg_giving_amt         16 175  0.98  0.53   0.99  0.03   2.47   2.44 0.04
## avg_giving_RT          17 175  6.70 18.04   4.74  2.13 234.63 232.50 1.36
## rHAP                   18 175  2.36  0.63   2.25  1.00   4.25   3.25 0.05
## rLAP                   19 175  2.87  0.79   2.75  1.00   5.00   4.00 0.06
## iHAP                   20 175  3.15  0.74   3.00  1.25   5.00   3.75 0.06
## iLAP                   21 175  3.95  0.70   4.00  2.00   5.00   3.00 0.05
```

```r
# European American
describe_all(data = data %>% dplyr::filter(cultureEA == 1), sensitivity_analyses)
```

```
## [1] "!is.na(subID)"
##                      vars   n  mean    sd median   min    max  range   se
## gender_nonfemale        1 103  0.39  0.49   0.00  0.00   1.00   1.00 0.05
## sexualpref_nonhetero    2 103  0.30  0.46   0.00  0.00   1.00   1.00 0.05
## ethn_nonmajority        3 103  0.00  0.00   0.00  0.00   0.00   0.00 0.00
## age                     4 103 26.19  6.61  25.00 18.00  66.00  48.00 0.65
## famises2_numbered       5 102  6.39  1.66   7.00  2.00   8.00   6.00 0.16
## selfses2_numbered       6 102  3.55  2.19   3.00  1.00   8.00   7.00 0.22
## tipi_ex                 7 103  3.59  1.71   3.50  1.00   7.00   6.00 0.17
## tipi_a                  8 103  4.83  1.21   4.50  2.00   7.00   5.00 0.12
## tipi_c                  9 103  5.39  1.17   5.50  2.50   7.00   4.50 0.12
## tipi_es                10 103  4.08  1.51   4.00  1.50   7.00   5.50 0.15
## tipi_o                 11 103  5.30  1.08   5.50  2.00   7.00   5.00 0.11
## iri_pt                 12 103  3.62  0.69   3.71  1.29   5.00   3.71 0.07
## iri_fs                 13 103  3.58  0.84   3.71  1.00   5.00   4.00 0.08
## iri_ec                 14 103  3.82  0.72   3.86  1.86   5.00   3.14 0.07
## iri_pd                 15 103  2.72  0.85   2.71  1.00   4.71   3.71 0.08
## avg_giving_amt         16  77  1.02  0.59   1.06  0.06   2.47   2.42 0.07
## avg_giving_RT          17 103  9.27 28.64   4.14  2.05 234.63 232.58 2.82
## rHAP                   18 103  2.42  0.76   2.25  1.25   5.00   3.75 0.07
## rLAP                   19 103  2.61  0.78   2.50  1.00   5.00   4.00 0.08
## iHAP                   20 103  3.32  0.82   3.25  1.00   5.00   4.00 0.08
## iLAP                   21 103  3.90  0.74   4.00  2.00   5.00   3.00 0.07
## [1] "exclude_age != 1"
##                      vars  n  mean    sd median   min    max  range   se
## gender_nonfemale        1 90  0.39  0.49   0.00  0.00   1.00   1.00 0.05
## sexualpref_nonhetero    2 90  0.31  0.47   0.00  0.00   1.00   1.00 0.05
## ethn_nonmajority        3 90  0.00  0.00   0.00  0.00   0.00   0.00 0.00
## age                     4 90 24.47  3.32  24.50 18.00  30.00  12.00 0.35
## famises2_numbered       5 89  6.44  1.64   7.00  2.00   8.00   6.00 0.17
## selfses2_numbered       6 89  3.34  2.14   3.00  1.00   8.00   7.00 0.23
## tipi_ex                 7 90  3.52  1.69   3.50  1.00   7.00   6.00 0.18
## tipi_a                  8 90  4.84  1.20   4.50  2.00   7.00   5.00 0.13
## tipi_c                  9 90  5.45  1.17   5.50  2.50   7.00   4.50 0.12
## tipi_es                10 90  4.07  1.52   4.00  1.50   7.00   5.50 0.16
## tipi_o                 11 90  5.25  1.06   5.50  2.00   7.00   5.00 0.11
## iri_pt                 12 90  3.63  0.69   3.71  1.29   5.00   3.71 0.07
## iri_fs                 13 90  3.60  0.86   3.71  1.00   5.00   4.00 0.09
## iri_ec                 14 90  3.83  0.74   3.86  1.86   5.00   3.14 0.08
## iri_pd                 15 90  2.76  0.84   2.71  1.00   4.71   3.71 0.09
## avg_giving_amt         16 69  1.03  0.59   1.06  0.06   2.47   2.42 0.07
## avg_giving_RT          17 90  7.90 25.69   4.15  2.05 234.63 232.58 2.71
## rHAP                   18 90  2.37  0.71   2.25  1.25   5.00   3.75 0.07
## rLAP                   19 90  2.60  0.81   2.50  1.00   5.00   4.00 0.09
## iHAP                   20 90  3.32  0.81   3.25  1.50   5.00   3.50 0.09
## iLAP                   21 90  3.91  0.76   4.00  2.00   5.00   3.00 0.08
## [1] "exclude_sd != 1"
##                      vars  n  mean    sd median   min    max  range   se
## gender_nonfemale        1 77  0.39  0.49   0.00  0.00   1.00   1.00 0.06
## sexualpref_nonhetero    2 77  0.35  0.48   0.00  0.00   1.00   1.00 0.05
```

```
## ethn_nonmajority        3 77  0.00  0.00   0.00  0.00   0.00   0.00 0.00
## age                     4 77 25.17  4.40  25.00 18.00  43.00  25.00 0.50
## famises2_numbered       5 76  6.41  1.59   7.00  2.00   8.00   6.00 0.18
## selfses2_numbered       6 76  3.50  2.09   3.00  1.00   8.00   7.00 0.24
## tipi_ex                 7 77  3.70  1.54   4.00  1.00   6.50   5.50 0.18
## tipi_a                  8 77  4.86  1.19   4.50  2.00   7.00   5.00 0.14
## tipi_c                  9 77  5.43  1.15   5.50  2.50   7.00   4.50 0.13
## tipi_es                10 77  4.03  1.46   4.00  1.50   7.00   5.50 0.17
## tipi_o                 11 77  5.31  1.05   5.50  2.00   7.00   5.00 0.12
## iri_pt                 12 77  3.64  0.68   3.71  1.29   4.86   3.57 0.08
## iri_fs                 13 77  3.60  0.80   3.57  1.00   5.00   4.00 0.09
## iri_ec                 14 77  3.84  0.69   3.86  1.86   5.00   3.14 0.08
## iri_pd                 15 77  2.71  0.81   2.71  1.00   4.71   3.71 0.09
## avg_giving_amt         16 77  1.02  0.59   1.06  0.06   2.47   2.42 0.07
## avg_giving_RT          17 77  7.63 26.27   4.14  2.13 234.63 232.50 2.99
## rHAP                   18 77  2.47  0.72   2.25  1.25   4.25   3.00 0.08
## rLAP                   19 77  2.65  0.75   2.50  1.00   4.50   3.50 0.09
## iHAP                   20 77  3.36  0.74   3.50  2.00   5.00   3.00 0.08
## iLAP                   21 77  3.91  0.71   4.00  2.25   5.00   2.75 0.08
## [1] "exclude_age != 1 & exclude_sd != 1"
##                      vars  n  mean    sd median   min    max  range   se
## gender_nonfemale        1 69  0.39  0.49   0.00  0.00   1.00   1.00 0.06
## sexualpref_nonhetero    2 69  0.35  0.48   0.00  0.00   1.00   1.00 0.06
## ethn_nonmajority        3 69  0.00  0.00   0.00  0.00   0.00   0.00 0.00
## age                     4 69 24.16  3.18  24.00 18.00  30.00  12.00 0.38
## famises2_numbered       5 68  6.44  1.57   7.00  2.00   8.00   6.00 0.19
## selfses2_numbered       6 68  3.28  2.04   3.00  1.00   8.00   7.00 0.25
## tipi_ex                 7 69  3.62  1.56   4.00  1.00   6.50   5.50 0.19
## tipi_a                  8 69  4.91  1.22   4.50  2.00   7.00   5.00 0.15
## tipi_c                  9 69  5.49  1.16   5.50  2.50   7.00   4.50 0.14
## tipi_es                10 69  4.04  1.50   4.00  1.50   7.00   5.50 0.18
## tipi_o                 11 69  5.28  1.06   5.50  2.00   7.00   5.00 0.13
## iri_pt                 12 69  3.67  0.66   3.71  1.29   4.86   3.57 0.08
## iri_fs                 13 69  3.61  0.83   3.71  1.00   5.00   4.00 0.10
## iri_ec                 14 69  3.86  0.71   3.86  1.86   5.00   3.14 0.09
## iri_pd                 15 69  2.73  0.83   2.71  1.00   4.71   3.71 0.10
## avg_giving_amt         16 69  1.03  0.59   1.06  0.06   2.47   2.42 0.07
## avg_giving_RT          17 69  7.85 27.74   4.16  2.13 234.63 232.50 3.34
## rHAP                   18 69  2.37  0.65   2.25  1.25   4.25   3.00 0.08
## rLAP                   19 69  2.64  0.77   2.50  1.00   4.50   3.50 0.09
## iHAP                   20 69  3.33  0.74   3.25  2.00   5.00   3.00 0.09
## iLAP                   21 69  3.89  0.73   4.00  2.25   5.00   2.75 0.09
```

```r
# Taiwanese
describe_all(data = data %>% dplyr::filter(cultureEA == 0), sensitivity_analyses)
```

```
## [1] "!is.na(subID)"
##                      vars   n  mean   sd median   min   max range   se
## gender_nonfemale        1 117  0.42 0.50   0.00  0.00  1.00  1.00 0.05
## sexualpref_nonhetero    2 117  0.17 0.38   0.00  0.00  1.00  1.00 0.03
## ethn_nonmajority        3 117  0.10 0.30   0.00  0.00  1.00  1.00 0.03
## age                     4 116 20.94 2.71  20.00 18.00 30.00 12.00 0.25
## famises2_numbered       5 117  3.61 1.65   4.00  1.00  8.00  7.00 0.15
## selfses2_numbered       6 114  1.07 0.29   1.00  1.00  3.00  2.00 0.03
## tipi_ex                 7 117  3.78 1.63   4.00  1.00  7.00  6.00 0.15
## tipi_a                  8 117  4.88 1.19   5.00  2.00  7.00  5.00 0.11
## tipi_c                  9 117  4.47 1.27   4.50  2.00  7.00  5.00 0.12
## tipi_es                10 117  3.84 1.26   4.00  1.00  7.00  6.00 0.12
## tipi_o                 11 117  4.75 1.12   5.00  1.50  7.00  5.50 0.10
## iri_pt                 12 117  3.54 0.67   3.43  1.57  5.00  3.43 0.06
## iri_fs                 13 117  3.57 0.76   3.57  1.43  5.00  3.57 0.07
## iri_ec                 14 117  3.50 0.74   3.57  1.29  5.00  3.71 0.07
## iri_pd                 15 117  3.10 0.67   3.14  1.43  4.57  3.14 0.06
## avg_giving_amt         16 107  0.96 0.48   0.99  0.03  2.36  2.33 0.05
## avg_giving_RT          17 117  5.79 6.02   4.96  2.34 66.00 63.66 0.56
## rHAP                   18 117  2.31 0.62   2.25  1.00  4.00  3.00 0.06
## rLAP                   19 117  3.02 0.78   3.00  1.25  5.00  3.75 0.07
## iHAP                   20 117  2.99 0.71   3.00  1.25  5.00  3.75 0.07
## iLAP                   21 117  3.99 0.68   4.00  2.00  5.00  3.00 0.06
## [1] "exclude_age != 1"
##                      vars   n  mean   sd median   min   max range   se
## gender_nonfemale        1 116  0.41 0.49   0.00  0.00  1.00  1.00 0.05
## sexualpref_nonhetero    2 116  0.17 0.38   0.00  0.00  1.00  1.00 0.04
## ethn_nonmajority        3 116  0.10 0.31   0.00  0.00  1.00  1.00 0.03
## age                     4 116 20.94 2.71  20.00 18.00 30.00 12.00 0.25
## famises2_numbered       5 116  3.60 1.66   4.00  1.00  8.00  7.00 0.15
## selfses2_numbered       6 113  1.07 0.29   1.00  1.00  3.00  2.00 0.03
## tipi_ex                 7 116  3.78 1.64   4.00  1.00  7.00  6.00 0.15
## tipi_a                  8 116  4.88 1.20   5.00  2.00  7.00  5.00 0.11
## tipi_c                  9 116  4.48 1.28   4.50  2.00  7.00  5.00 0.12
## tipi_es                10 116  3.84 1.26   3.75  1.00  7.00  6.00 0.12
## tipi_o                 11 116  4.75 1.12   5.00  1.50  7.00  5.50 0.10
## iri_pt                 12 116  3.55 0.68   3.50  1.57  5.00  3.43 0.06
## iri_fs                 13 116  3.57 0.76   3.57  1.43  5.00  3.57 0.07
## iri_ec                 14 116  3.50 0.74   3.64  1.29  5.00  3.71 0.07
## iri_pd                 15 116  3.10 0.67   3.14  1.43  4.57  3.14 0.06
## avg_giving_amt         16 106  0.95 0.48   0.97  0.03  2.36  2.33 0.05
## avg_giving_RT          17 116  5.80 6.05   4.93  2.34 66.00 63.66 0.56
## rHAP                   18 116  2.31 0.62   2.25  1.00  4.00  3.00 0.06
## rLAP                   19 116  3.02 0.78   3.00  1.25  5.00  3.75 0.07
## iHAP                   20 116  3.00 0.71   3.00  1.25  5.00  3.75 0.07
## iLAP                   21 116  4.00 0.68   4.00  2.00  5.00  3.00 0.06
## [1] "exclude_sd != 1"
##                      vars   n  mean   sd median   min   max range   se
## gender_nonfemale        1 107  0.41 0.49   0.00  0.00  1.00  1.00 0.05
## sexualpref_nonhetero    2 107  0.17 0.38   0.00  0.00  1.00  1.00 0.04
```

```
## ethn_nonmajority        3 107  0.09 0.29   0.00  0.00  1.00  1.00 0.03
## age                     4 106 20.86 2.75  20.00 18.00 30.00 12.00 0.27
## famises2_numbered       5 107  3.47 1.54   4.00  1.00  8.00  7.00 0.15
## selfses2_numbered       6 104  1.07 0.29   1.00  1.00  3.00  2.00 0.03
## tipi_ex                 7 107  3.81 1.64   4.00  1.00  7.00  6.00 0.16
## tipi_a                  8 107  4.84 1.20   4.50  2.00  7.00  5.00 0.12
## tipi_c                  9 107  4.41 1.22   4.50  2.00  7.00  5.00 0.12
## tipi_es                10 107  3.80 1.20   3.50  1.00  7.00  6.00 0.12
## tipi_o                 11 107  4.77 1.07   5.00  1.50  7.00  5.50 0.10
## iri_pt                 12 107  3.55 0.66   3.57  1.57  5.00  3.43 0.06
## iri_fs                 13 107  3.59 0.75   3.57  1.43  5.00  3.57 0.07
## iri_ec                 14 107  3.53 0.73   3.71  1.29  5.00  3.71 0.07
## iri_pd                 15 107  3.10 0.62   3.14  1.57  4.43  2.86 0.06
## avg_giving_amt         16 107  0.96 0.48   0.99  0.03  2.36  2.33 0.05
## avg_giving_RT          17 107  5.95 6.25   5.05  2.44 66.00 63.56 0.60
## rHAP                   18 107  2.35 0.62   2.25  1.00  4.00  3.00 0.06
## rLAP                   19 107  3.01 0.76   3.00  1.25  5.00  3.75 0.07
## iHAP                   20 107  3.03 0.71   3.00  1.25  5.00  3.75 0.07
## iLAP                   21 107  3.98 0.69   4.00  2.00  5.00  3.00 0.07
## [1] "exclude_age != 1 & exclude_sd != 1"
##                      vars   n  mean   sd median   min   max range   se
## gender_nonfemale        1 106  0.41 0.49   0.00  0.00  1.00  1.00 0.05
## sexualpref_nonhetero    2 106  0.17 0.38   0.00  0.00  1.00  1.00 0.04
## ethn_nonmajority        3 106  0.09 0.29   0.00  0.00  1.00  1.00 0.03
## age                     4 106 20.86 2.75  20.00 18.00 30.00 12.00 0.27
## famises2_numbered       5 106  3.46 1.55   4.00  1.00  8.00  7.00 0.15
## selfses2_numbered       6 103  1.07 0.29   1.00  1.00  3.00  2.00 0.03
## tipi_ex                 7 106  3.82 1.65   4.00  1.00  7.00  6.00 0.16
## tipi_a                  8 106  4.84 1.20   4.75  2.00  7.00  5.00 0.12
## tipi_c                  9 106  4.42 1.23   4.50  2.00  7.00  5.00 0.12
## tipi_es                10 106  3.79 1.20   3.50  1.00  7.00  6.00 0.12
## tipi_o                 11 106  4.77 1.08   5.00  1.50  7.00  5.50 0.10
## iri_pt                 12 106  3.56 0.67   3.57  1.57  5.00  3.43 0.06
## iri_fs                 13 106  3.60 0.75   3.64  1.43  5.00  3.57 0.07
## iri_ec                 14 106  3.53 0.73   3.71  1.29  5.00  3.71 0.07
## iri_pd                 15 106  3.10 0.62   3.14  1.57  4.43  2.86 0.06
## avg_giving_amt         16 106  0.95 0.48   0.97  0.03  2.36  2.33 0.05
## avg_giving_RT          17 106  5.96 6.28   5.06  2.44 66.00 63.56 0.61
## rHAP                   18 106  2.35 0.62   2.25  1.00  4.00  3.00 0.06
## rLAP                   19 106  3.02 0.76   3.00  1.25  5.00  3.75 0.07
## iHAP                   20 106  3.03 0.71   3.00  1.25  5.00  3.75 0.07
## iLAP                   21 106  3.99 0.68   4.00  2.00  5.00  3.00 0.07
```